

PERSONAL DIARY MANAGEMENT SYSTEM

1. Title Page

Project Title: Personal Diary Management System

Course: Programming in C

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Academic Year: 2025-26

2. Abstract

This project implements a Personal Diary Management System using the C programming language. The primary objective of this system is to allow users to digitally store and manage diary entries. The application provides essential operations including adding new diary entries, viewing saved entries, searching specific entries, and deleting existing entries. Data storage is handled using text files, enabling persistent storage even after the program terminates. The project demonstrates the use of key C programming concepts such as file handling, structures, loops, string manipulation, and modular programming. The final program improves organization and efficiency compared to manual diary maintenance.

3. Problem Definition

A personal diary is a record of events, thoughts, and reminders written on a daily basis. Traditionally, diaries are handwritten, which can lead to problems such as loss, damage, disorganization, and difficulty retrieving previous entries.

The goal of this project is to develop a digital method for creating and managing diary entries using C programming. The system must:

- Allow entry creation with a date, title, and content
 - Store entries permanently in a text file
 - Display all stored entries
 - Search entries using keywords
 - Delete entries when needed
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4. System Design

4.1 Algorithm

Main Menu Algorithm:

1. Start
2. Display menu options
3. Accept user input
4. Based on the choice:
 - Add new entry
 - View entries
 - Search entry
 - Delete entry
 - Exit
5. Return to menu until exit is selected

Add Entry Algorithm:

1. Open file in append mode
2. Take user inputs (date, title, content)
3. Store in file in formatted structure
4. Close file

View Entry Algorithm:

1. Open file in read mode
2. Read all lines and display formatted output
3. Close file

Search Entry Algorithm:

1. Open file in read mode
2. Read each entry and compare with search keyword
3. If match found, display entry
4. If no match, display not found message

Delete Entry Algorithm:

1. Open main file and temporary file

2. Read line by line
 3. Copy only non-matching entries to temporary file
 4. Replace original file with temporary file
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5. Implementation Details

Programming Concepts Used:

- C File Handling (fopen(), fprintf(), fgets(), remove(), rename())
- Structures
- Functions
- Loops and Conditionals
- String Manipulation

Code Snippet Example:

```
struct Diary {  
char date[15];  
char title[50];  
char content[500];  
};  
  
void addEntry() {  
struct Diary d;  
FILE *fp = fopen(FILENAME, "a");  
(Full code available inside /src folder)
```

6. Conclusion & Future Work

The Personal Diary Management System successfully performs core diary functions like storing, viewing, searching, and deleting entries. The project enhanced understanding of file operations and modular coding in C.

Future Enhancements:

- Password protection
- Sorting entries by date

- Exporting entries to PDF
- Multi-user support
- Graphical User Interface

```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

Enter Title: day 1
Write your content:
my day was great

rflà Entry added successfully!

==== PERSONAL DIARY APPLICATION ====
1. Add New Entry
2. View All Entries
3. Search Entry
4. Delete Entry
5. Exit
Enter your choice: 1

Enter Date (DD/MM/YYYY): 2-12-2024
Enter Title: day2
Write your content:
how was yours?

rflà Entry added successfully!

==== PERSONAL DIARY APPLICATION ====
1. Add New Entry
2. View All Entries
3. Search Entry
4. Delete Entry
5. Exit
Enter your choice: 2

==== ALL DIARY ENTRIES ====

Date: AppData\Local\
Title: Temp\" ; if ($?) { gcc tempCodeRunnerFile.c -o te
Content: mpCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
=====
```